# GIVE SMILE NGO & FOUNDATION

Platform Security & Data Protection Technical Specification

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## 1. Executive Summary

This document specifies the technical security architecture, system safeguards, and data protection protocols implemented across the **Give Smile NGO** web platform. The platform is engineered with a dual-tier architecture combining a **React 18 + Vite** frontend and a **Spring Boot 3.2.3 (Java 17)** REST backend. The system ensures complete data protection, eliminates financial payment tampering, prevents injection vectors, and safeguards user privacy.

## 2. Backend Security Architecture (Spring Boot & Java 17)

• **SQL Injection Immunity:** All database access uses **Spring Data JPA & Hibernate ORM** parameterized queries, preventing direct SQL concatenation and insulating the database against SQL injection vectors.

• **Anti-Forgery Cryptographic Receipts:** Donation pledges generate unique, cryptographically random audit codes combining system timestamps and 128-bit UUID fragments (e.g., GSF-TIMESTAMP-UUID), rendering receipt tampering and enumeration attacks impossible.

• **CORS Request Control:** REST controllers (/api/donations, /api/contact, /api/volunteers) enforce explicit origin policies, guarding backend APIs against unauthorized cross-domain exploitation.

• **Payload Flooding & DoS Protection:** Configured Spring Servlet multipart limits (max-file-size=50MB, max-request-size=50MB) in application properties, safeguarding system memory buffers against Denial of Service (DoS) attacks.

• **Off-Site Audit Relay:** Asynchronous notification services dispatch real-time logs to contact@givesmilefoundation.org upon any form submission for off-site accountability.

## 3. Frontend Security Architecture (React 18 & Vite)

• **Uneditable Dynamic Payment QR Security:** Embedded payment payload strings (upi://pay?pa=...&={AMOUNT}&cu;=INR) dynamically pre-fill the selected contribution and lock input parameters in an uneditable UI state, ensuring 100% protection against payment amount fraud.

• **Cross-Site Scripting (XSS) Mitigation:** React JSX auto-escaping guarantees all dynamic user inputs (names, messages, comments) render exclusively as plain-text DOM nodes, mitigating stored and reflected XSS attacks.

• **Strict Route Allow-List Navigation:** Enforced hash navigation white-lists in App.jsx (VALID_PAGES). Malicious fragments or unknown route requests automatically default to home, eliminating Open Redirect risks.

• **Input Validation & Regex Filtering:** Client-side regular expression pattern validation filters telephone numbers (10 digits), email formats, and positive numeric amounts prior to API transmission.

• **Production Bundle Shielding:** Vite/Rollup production compilation minifies code structures and strips debug maps, safeguarding internal business logic.

## 4. Security Safeguard Verification Matrix

| Security Domain | Implementation Mechanism | Target Risk Mitigated | Status |
|---|---|---|---|
| Database Access | Spring Data JPA / Hibernate ORM | SQL Injection & Data Tampering | **VERIFIED** |
| Payment Integrity | Dynamic UPI Payload + Locked UI | Payment Amount Alteration / Fraud | **VERIFIED** |
| Receipt Audit | UUID + System Epoch Cryptography | Receipt Forgery & Enumeration | **VERIFIED** |

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|---|---|---|---|
| Client Rendering | React JSX Auto-Escaping | DOM-based & Reflected XSS | **VERIFIED** |
| Route Navigation | App.jsx Strict Page Allow-List | Open Redirects & Fragment Injection | **VERIFIED** |
| Buffer Management | 50MB Spring Servlet Multipart Caps | Buffer Overflow & Memory DoS | **VERIFIED** |